Question #: 1

An investigator studying nutrition asks participants to keep a food and exercise diary for a month. An entry from a 43-year-old man weighing 73 kg (160 lb) shows that he consumed 300 g of carbohydrates, 80 g of proteins, 90 g of fats, 20 g of ethanol, and 25 g of dietary fiber on the previous day. His daily energy expenditure was approximately 2400 Calories. Which of the following would be the best conclusion from these data?

A. He has to decrease the dietary fiber content for optimal health
B. To maintain the present weight, he needs to consume an additional 400 Calories
C. He is consuming 200 Calories more than his requirement
D. He has to reduce the intake of dietary protein to maintain nitrogen balance
E. Dietary fats are providing about 33% of his total calorie intake

Question #: 2

An investigator studying nutrition prepares meals containing amino acids of various compositions as the sole contribution of dietary protein. These meals are fed to a group of normal adult female volunteers who are 25-30 years old. A protein deficient in which of the following amino acids would most likely result in a negative nitrogen balance in these research subjects?

A. Cysteine
B. Leucine
C. Tyrosine
D. Asparagine
E. Glutamate

Question #: 3

A 25-year-old man comes to the physician for a follow-up examination. He says he gained 10-kg (22-lbs) during the past 3 years. He now weighs 90 kg (198 lb), his waist-to-hip ratio is 1.4; and BMI is 34 kg/m^2. Which of the following metabolic changes most likely occurred following this patient's change in weight?

A. Decrease in the size of adipocytes
B. Increase in the secretion of leptin
C. Activation of the hunger center
D. Increase in the secretion of ghrelin
E. Decrease in the number of adipocytes

Question #: 4

An investigator discovers an aberrant karyotype in an unaffected mother of a child who has an intellectual disability and dysmorphology. The karyotype of the mother shows a fusion of the long arms of two non-homologous acrocentric chromosomes and she has 45 total chromosomes. Which of the following best explains the appearance of this karyotype?

A. Reciprocal translocation
B. Robertsonian translocation
C. Meiosis I nondisjunction
D. Chromosome inversion
E. Autosomal triploidy
F. Imprinting error

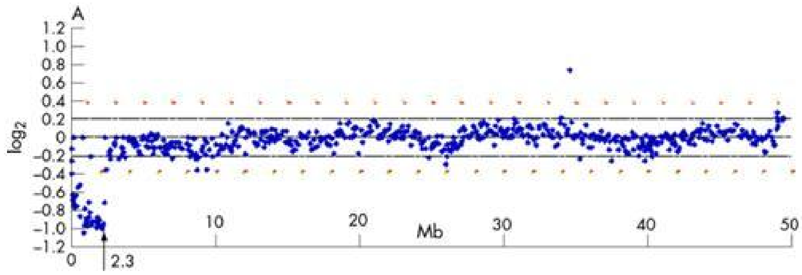
Question #: 5

A 11-year-old boy is brought to the physician because of development delay, intellectual disability, and facial anomalies. The child’s G-banded metaphase karyotype appeared normal, and showed no detectable chromosomal abnormalities including, inversions, insertions, deletions, or duplications. A different genetic showed an alteration on chromosome 5, as shown in the figure. The G-band karyotype did not detect the abnormality because it

A. Offers less resolution than array comparative genome hybridization
B. Does not offer information regarding gene expression
C. May only be used with chromosomes isolated from mitotic cells
D. Is not able to detect deletions on a chromosome

E. Relies on polymerase chain reaction technology
F. Requires the hybridization of a radioactively labeled probe

Attachment:



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Question #: 6

A 13-year-old boy is brought to the physician by his adoptive parents for a follow-up examination because of a lifetime history of mild learning disabilities, mild behavior problems, and acne. Standard karyotype of 40 metaphase cells shows that the boy has a mosaic karyotype consisting of 46,XY, (30 cells) and 47,XY+16 (10) cells. Which of the following is most likely responsible for this patient's mixed karyotype?

A. Maternal nondisjunction at meiosis I
B. Maternal nondisjunction at meiosis II
C. Mitotic nondisjunction after fertilization
D. Paternal nondisjunction at meiosis II
E. Paternal nondisjunction at meiosis I

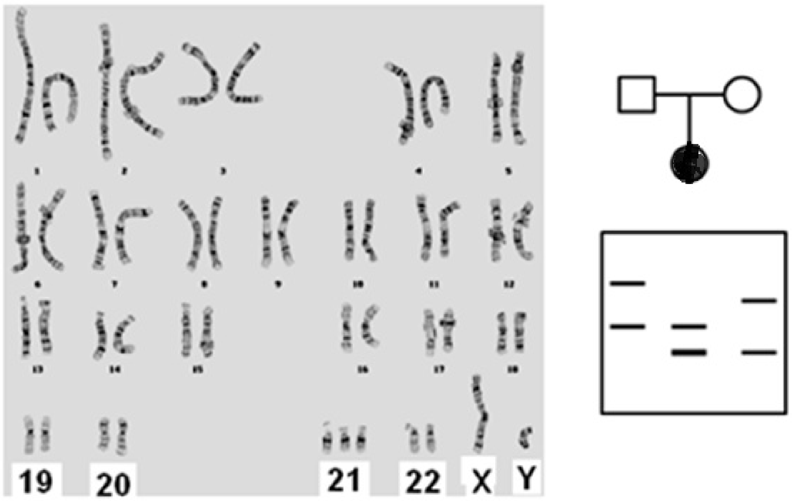
Question #: 7

A 32-year-old woman comes to the physician for follow up after giving birth to a child with dysmorphic facial features that included a depressed nasal bridge and upturned palpebral fissures. The karyotype of the child and polymorphic marker analysis of the family are shown. Based on these data, which of the following most likely caused the aneuploidy condition in the child?

A. Robertsonian translocation involving chromosome 21
B. Meiosis I nondisjunction in the mother
C. Meiosis I nondisjunction in the father

D. Meiosis II nondisjunction in the mother
E. Meiosis II nondisjunction in the father

Attachment:



ExamSoft ID 21843 post mt online Q karyotype trisomy 21 and markers.jpg

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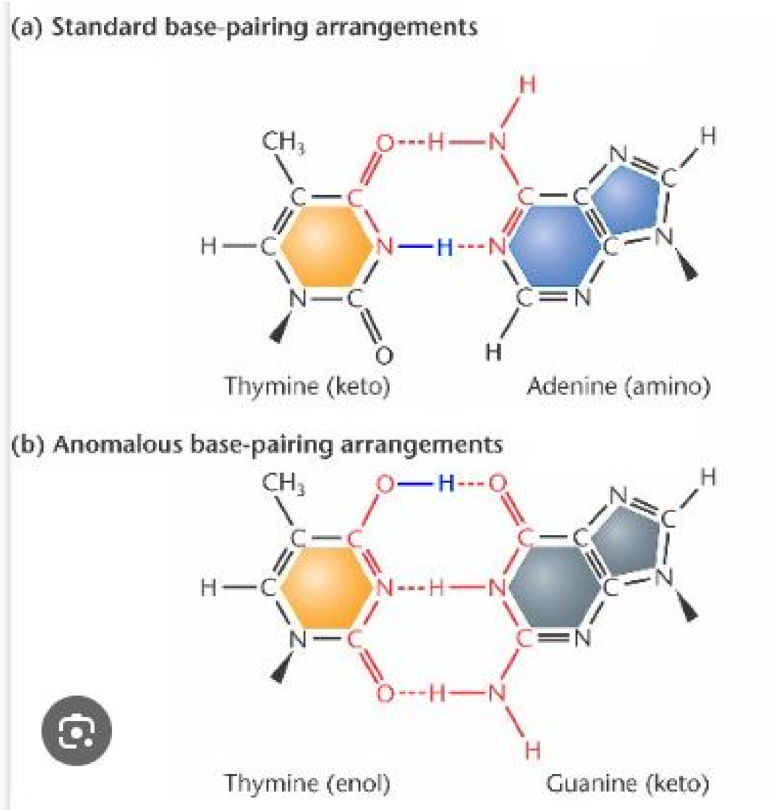
Question #: 8

A scientist studying genome stability creates an assay to detect DNA repair by the proof-reading mechanism. The assay measures repair of mis-incorporated bases that were caused by tautomeric shifts of thymine as shown in the attached figure. In which of the following stage of the cell cycle did this repair most likely occur?

A. G1 phase
B. S phase
C. G2 phase
D. M phase
E. G0 phase

Attachment:



Tautomer.JPG

Question #: 9

A 16-year-old girl is brought to the physician by her mother because of a severe sunburn. Her sunburn is due to 3 hours of exposure to ultraviolet light, a component of the electromagnetic spectrum of sunshine. Because of this exposure, DNA damage accumulates in her skin cells. Which of the following repair systems is most likely responsible for the correction of this damaged DNA before replication?

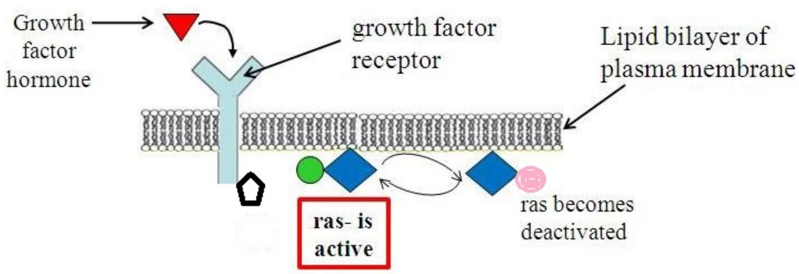
A. Mismatch repair
B. Base excision repair
C. Non-homologous end joining
D. Nucleotide excision repair
E. Proofreading

Question #: 10

A researcher studying cancer is investigating the signal transduction pathway that cells use to regulate cell division. A diagram of the major cell proliferative pathway shown. Which of the following must occur to cause a normally functioning Ras protein to become active in the context of this pathway?

A. Ras mus be phosphorylated on a tyrosine
B. Ras must disengage from the membrane
C. Ras must be bound to GDP
D. Receptor must become dephosphorylated
E. Growth factor hormone must bind to receptor

Attachment:



RAS cartoon_a.jpg

Question #: 11

A 9-month-old boy is brought to the opthamologist by his parents after observing strange white reflections in both eyes in all photographs taken with a flash in the last month. Physical examinition with an opthalmoscope shows a 2-milimeter white fluffy retinal mass in both eyes. Which of the following cellular events is most likely asociated with the formation of the tumors in this patient?

A. Activation of p53
B. Loss of 3' to 5' proof reading of DNA polymerase
C. Activation of the G2/M checkpoint
D. Release of cytochrome c from the mitochondria
E. Promotion of the G1/S transition

Attachment:

Rb.JPG

Question #: 12

A 33-year-old scientist studying DNA repair subjects an in-vitro culture of human epithelial cells to x-ray radiation. The ionizing radiation induces DNA strand breakage. Which of the following events would most likely occur in these cells as a result of this damage?

A. The BCL-2 gene will be expressed at high levels
B. The perforin/granzyme apoptotic pathway will be activated
C. The p53 protein will be inactive and unphosphorylated
D. The amount of BAX protein will be elevated in the cell
E. The survival pathway will be activated

Question #: 13

A 12-year-old girl is brought to the physician by her parents for follow-up because of the progressive appearance of neurofibromas. Physical examination shows she now has more than 200 distinct formations of these neoplasms, and Lisch nodules on the iris of yer eyes. Her father and paternal grandmother had a similar condition. Which of the following is the most probable normal function of the gene that is mutated in this patient?

A. DNA repair
B. Proto-oncogene
C. Tumor suppressor
D. Apoptosis regulator
E. Cell cycle checkpoint regulator

Question #: 14

A researcher studying colon cancer finds that 2 –5% of colon cancer cases are heritable. One familial type is called HNPCC (hereditary non-polyposis colon cancer). Compilation of the genomic structure and mutation profile of cells obtained from 100 resected HNPCC tumors shows a specific pathognomonic signature. Which of the following best describes the function of the pathway that, when mutationally inactivated, leads to the development of this type of cancer?

A. Activation of G2/M checkpoint
B. Signaling through APC
C. DNA mismatch repair
D. Nucleotide excision repair
E. Initiation of apoptosis

Question #: 15

A researcher studying cancer in an in vitro model system, uses site directed mutagenesis to alter one allele of a gene in a cell. The missense base-pair change leads to an amino acid substitution in the encoded protein. In response to the mutation, the cell line is found to attain the properties of oncogenic transformation and unregulated proliferation. Which of the following genes is most likely to have been mutated by the researcher?

A. *RB*
B. *RAS*
C. *XPA*
D. *MSH2*
E. *HTT* (the gene associated with Huntington disease)
F. *HEXA*

G. *MECP2*

Question #: 16

An 8-year-old boy with progressive swelling of the jaw is brought to the physician by his parents. A biopsy is performed and FISH analysis of the tissue shows a translocation between chromosomes 8 and 14. Proteomic analysis of the tissue shows the elevated expression of an oncoprotein. Which of the following best describes the function of this protein?It

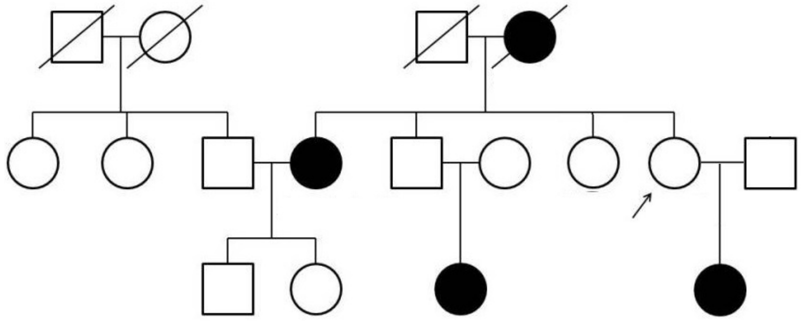
A. It has tyrosine kinase activity
B. It functions as a growth factor receptor
C. It activated the G1-S checkpoint
D. It is required for DNA repair
E. It is a transcription factor
F. It activated apoptosis

Question #: 17

A 72-year-old woman comes to the physician for a follow-up appointment to discuss results of her mammogram. Her 41-year-old daughter was recently diagnosed with breast cancer. Results of the mammogram show no evidence of breast cancer. A detailed family history is obtained, and the resulting pedigree of affected family members is shown. The patient is indicated by the arrow. Genetic testing shows that the patient, her daughter and other affected family members have the same pathogenic germline mutation in BRCA2. Which of the following best describes the genetic principle illustrated in this patient?

A. Allelic heterogeneity
B. Incomplete penetrance
C. Heteroplasmy
D. Anticipation
E. Variable expressivity

Attachment:



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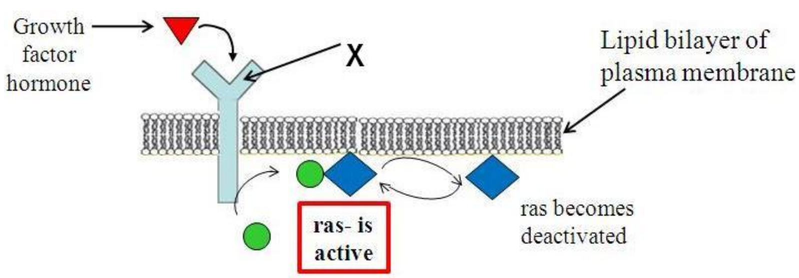
Question #: 18

A researcher studying cancer obtains a biopsy from a 69-year-old man to study the genetic driver mutations involved in the development of the metastatic potential of the neoplasm. Next-generation sequencing shows that the cancer cells have a mutation in the gene which encodes the protein labeled "X" as shown in the figure. Which of the following best describes this protein?

A. Small monomeric G-protein
B. Tyrosine kinase
C. Transcription factor
D. Steroid binding protein
E. IgG Antibody
F. DNA helicase enzyme

Attachment:



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Question #: 19

A 12-year-old boy is brought to the physician for an annual health maintenance examination. He has intellectual disability, short stature, a depressed nasal bridge, upslanting palpebral fissures, epicanthal folds, single palmar creases, and an exaggerated sandal gap on both feet. He was born with a congenital heart defect which was surgically repaired. Genetic testing when he was an infant showed that the disorder was caused by a nondisjunction event during oogenesis.

Which of the following syndromes does this patient most likely have?

A. Cri-du-chat syndrome
B. Klinefelter syndrome
C. Down syndrome
D. Wolf-Hirschhorn syndrome
E. Angelman syndrome

Question #: 20

A 70-year-old man is brought to the physician for a follow-up examiantion. He has a 6-month history of chest discomfort, trouble breathing, a worsening cough, and blood in his sputum. A CT scan show small lesions in his lungs, sputum cytology shows the presence of lung cancer cells and a diagnosis of stage 2 non-small cell lung cancer is made. The patient agrees to give a blood sample for researching biomarkers of non-small cell lung cancer before and after 2-months of chemotherapy. Tests show that the patient initially has 5-times lower concentrations of cytochrome c in his serum versus control patients. After 2-months of chemotherapy, his serum cytochrome c levels are 10-times higher than control levels. Which of the following cellular events is most likely to cause the release of this component from its normal cellular location?

A. Cleavage of procaspases
B. Formation of the apoptosome
C. Blebbing of the plasma membrane
D. BAX is inserted into the mitochondrial membrane
E. Increased expression of BCL2

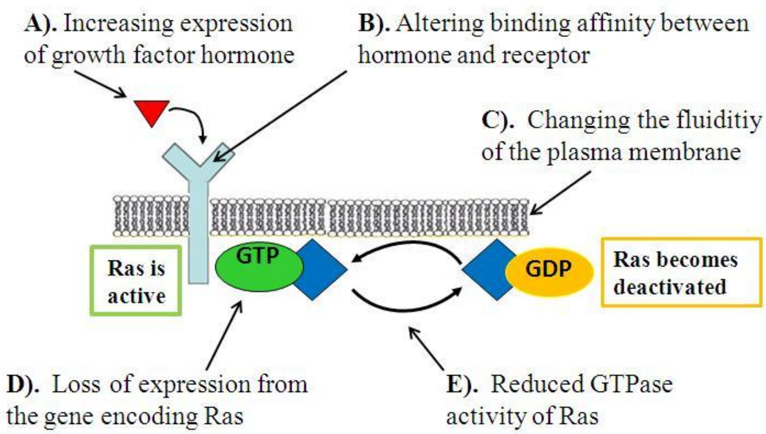
Question #: 21

An 8-year-old girl is brought to the physician by her legal guardian for a follow-up examination after a genetic test confirms a diagnosis of neurofibromatosis. A figure depicting the pathway affected by this pathogenic genetic variant is shown. Which of the following functions indicated in the figure best provides a molecular explanation of the disorder in this girl?

A. A

B. B
C. C
D. D
E. E

Attachment:



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Question #: 22

A 5-year-old girl is brought to the physician by her foster parents because of progressive impairment of muscular coordination and a multitude of surface blood vessels particularly in the eyes. Analysis of the sequence of ATM shows that the girl has compound heterozygosity for loss of function mutations in this gene. Which of the following activities is most likely to be directly compromised as a result of the genetic mutations in this patient?

A. DNA helicase activity
B. Detection of double strand breaks
C. Repair of pyrimidine dimers
D. Post replication mismatch repair
E. Transcription of growth promoting genes

Question #: 23

A researcher is studying genetic factors that influence the risk of multiple sclerosis. This investigator is attempting to associate the allelic variants in patients with the disorder, to compare against normal controls (individuals without the condition). Which of the following techniques was most likely used to collect the data in this analysis?

A. FISH analysis
B. Northern analysis
C. SNP chip
D. G-band karyotype
E. ASO blot
F. Southern blot
G. Western blot

Question #: 24

A researcher is attempting to catalog all genetic disorders associated with human mortality and morbidity. Which type of the heritable conditions contribute the most to the disease load in the typical human population?

A. Chromosomal aberrations
B. Autosomal dominant disorders
C. Multifactorial disorders
D. Digenic disorders
E. Autosomal recessive disorders
F. Mitochondrial disorders

Question #: 25

A researcher studying non-syndromic congenital heart defects (CHD) finds that the population incidence of this multifactorial developmental defect is approximately 1 in 100 live births. Comparisons are made to estimate recurrence risk depending on the affected relative. Which of the following best explains an aspect of the inheritance of this disorder?

A. Recurrence risk will be higher in their male children then in their female children
B. Recurrence risk reduces dramatically as relatedness decreases
C. Recurrence risk will be the same as the population incidence
D. Recurrence risk will be one half of the population incidence
E. There is no risk to the children as this is a quantitative trait

F. Recurrence risk will be twice the population incidence

Question #: 26

A 32-year-old woman comes to the physician for a follow up consultation after being diagnosed with coronary artery disease. Research shows that compared to men, premenopausal women are typically at lower risk for this disorder because of the protective effects of estrogen. Which of the following best summarizes the recurrence risk prediction among the immediate members of this woman’s family?

A. Her first degree relatives are at population risk levels
B. One of her parents must have had the condition
C. Her male children are at an increased risk for the disorder
D. Her female children are at a reduced risk for the disorder
E. Each of her children has a ½ risk of inheriting the disorder

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Question #: 27

An investigator is studying the cardiovascular effects of cholinergic drugs. He injects a dose of methacholine intravenously in an anesthetized dog. The dog's heart rate decreases. Which of the following best explains this finding?

A. Activation of M1 muscarinic receptors on vascular smooth muscle.
B. Activation of M2 muscarinic receptors on cardiac tissue
C. Activation of M3 muscarinic receptors on vascular endothelial cells.
D. Activation of nicotinic receptors on autonomic ganglia
E. Activation of presynaptic M2 muscarinic receptors in adrenergic terminals.

Question #: 28

A pharmacologist is studying the cardiovascular actions of a novel drug, Drug X. He administers an intravenous infusion of Drug X to an anesthetized dog. The infusion evokes an increase in both blood pressure and mean arterial pressure. Which of the following mechanisms best explains this effect?

A. Direct myocardial depression through activation of β1 receptors
B. Direct myocardial depression through activation of M2 receptors
C. Decreased heart rate through activation of β1 receptors
D. Constriction of blood vessels through activation of α1 receptors
E. Dilation of blood vessels in skeletal muscle through activation of β2 receptors

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Question #: 29

A 21-year-old woman comes to the physician for a routine ophthalmologic examination. A solution of phenylephrine is instilled into her eyes to dilate her pupils prior to examination. Which of the following changes is most likely to occur in the smooth muscle cells of the patient's pupil in response to the instilled drug?

A. Decreased levels of cAMP
B. Enhanced GABAergic transmission
C. Increased levels of cGMP
D. Increased levels of IP3
E. Decreased levels of cGMP
F. Increased levels of cAMP
G. Decreased levels of IP3

Question #: 30

An investigator administers an intravenous infusion of low-dose epinephrine to an anesthetized dog. The infusion evokes a decrease in diastolic blood pressure. Which of the following mechanisms best explains this effect?

A. Direct myocardial depression through activation of beta1 receptors
B. Direct myocardial depression through activation of M2 receptors
C. Decreased heart rate through activation of beta1 receptors
D. Dilation of blood vessels of skin, mucosa and kidney through activation of alpha1 receptors
E. Dilation of blood vessels in skeletal muscle through activation of beta2 receptors